| INCH-POUND |

MIL-P-16077D(YD)
21 June 1991
SUPERSEDING
MIL-P-16077C(YD)
30 October 1985

MILITARY SPECIFICATION

PUMPS, CENTRIFUGAL, WATER CIRCULATING, ELECTRIC-MOTOR-DRIVEN

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers electric-motor-driven, centrifugal, water circulating pumps for use in hot water heating systems and service water distribution systems. The pumps have in-line connections to permit unsupported installation of the units in a straight run of piping.

1.2 Classification. Pumps are of the following types, sizes, and intended service, as specified (see 6.2):

Type I - Pump shaft in horizontal plane.
Type II - Pump shaft in vertical plane.

Sizes - 0.75, 1.00, 1.25, 1.50, 2.00, 2.50, 3.00 inches.

Service A - Hot water heating systems.
Service B - Service water distribution systems.

| Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. |

AMSC N/A

FSC 4320

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications. The following specifications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the specification (see 6.2).

SPECIFICATION

MILITARY

MIL-P-10603 - Pumps and Pumping Units, Centrifugal, Power-Driven, for Water; Packaging of.

(Unless otherwise indicated, copies of specifications are available from Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B 16.1 - Cast Iron Pipe Flanges and Flanged Fittings.
ASME B 16.24 - Bronze Pipe Flanges and Flanged Fittings Class 150 and 300.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The pumps covered by this document shall be the in-line, centrifugal, electric-motor-driven type. The motor shall be supported in a mounting bracket attached to the pump casing to form an integrated, self-contained unit. The size of the pump is determined by the diameter, in inches, of the discharge.

3.2 First article. When specified (see 6.2), the contractor shall furnish a pump for first article inspection and approval (see 4.2.1 and 6.6).

3.3 Standard commercial product. The pump shall, as a minimum, be in accordance with the requirements of this specification and may be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, may be included in the pump being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements of manufacturer's catalogs, or brochures, and represents the latest production model.

3.3.1 System of measurement. The dimensions used in this specification are not intended to preclude the use of metric system of measurement in the fabrication and production of the material, individual parts, and the finished product, provided form, fit, and function requirements are satisfied.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this document are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

3.5 Interchangeability. All pumps of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

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3.6 Performance. The pump shall be capable of meeting the capacity and total dynamic head requirements specified (see 6.2), based on pumping water at a temperature of 80 degrees Fahrenheit (^{o}F). Pumps rated at 1/2 horsepower (hp) or greater shall be capable of continuously delivering its rated capacity at a pump efficiency of not less than 50 percent. The pumps shall be capable of handling water at temperatures up to 225°F and shall be furnished complete in all details, including those not specifically mentioned herein. The pump shall be designed to withstand a hydrostatic pressure of not less than 1.5 times the maximum rated head of the pump. The pump design shall permit accessibility for maintenance and service in the field, and shall be such as to prevent conditions which may be hazardous to operating personnel.

3.7 Construction. The pumps shall include a motor, motor mounting bracket, and pump with inlet and outlet connections on opposite sides of the casing on a common centerline. Pumps shall be constructed to permit unsupported installation in a straight run of horizontal or vertical piping. For type I, the horizontal pump shall be connected to the motor by a shaft with a flexible coupling designed to minimize the transmission of motor vibrations to the pump casing and connected piping. For type II, the vertical pump shall be connected to the motor by a thrust-carrying solid extended motor shaft.

3.7.1 Pump casing. Unless otherwise specified (see 6.2), the pump casing and flange shall be made of close-grained cast iron for service A and brass or bronze for service B.

3.7.2 Impeller. Unless otherwise specified (see 6.2), the impeller shall be made of a suitable nonferrous material. When specified (see 6.2), the impeller shall be made of brass or bronze. The impeller shall be open, semi-enclosed, or enclosed type to meet the required performance. The impeller shall be mechanically balanced, and shall be keyed and locked or otherwise securely fastened to the pump shaft to prevent backing-off or loosening with change of rotation.

3.7.3 Pump shaft. Unless otherwise specified (see 6.2), the pump shaft shall be alloy steel or nickel alloy. The pump shall have oil lubricated sleeve bearing or grease-lubricated ball bearings. Sleeve bearings shall be lubricated by oil supplied either by a capillary material from an oil reservoir or centrifugally by a slinger affixed to the shaft rotating inside an oil sump chamber.

3.7.4 Connections. Direction of flow shall be indicated by permanent marking on the pump casing. Inlet and outlet connections shall be flanged and, unless otherwise specified (see 6.2), shall be furnished with threaded companion flanges. Flanges for sizes up to and including 1.50 inches shall be not less than two-bolt; flanges for sizes over 1.50 inches shall be not less than four-bolt. Flanges for sizes 2.50 inches and over shall conform to ASME B 16.1 and ASME B 16.24, as applicable.

3.7.5 Seal. A replaceable seal shall be provided between the pump chamber and the bearing housing to prevent leakage of water from the pump chamber to the bearing housing. Seals furnished shall be suitable for use at the temperatures and pressures specified herein.

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3.7.6 Motor. The motor shall be constructed and rated in accordance with the applicable provisions of NEMA MG 1. Unless otherwise specified (see 6.2 and 6.1.1), motors supporting loads in excess of 1/2 hp shall be a three-phase, continuous duty, 60 Hertz, 208-230/460 volt (V), with a drip-proof enclosure design, and operate in an ambient temperature of 104°F. For loads of 1/2 hp and less, single-phase motors having a nominal 115V rating may be used. The motor shall be furnished with built-in thermal overload protection and shall have a rated speed not to exceed 1,800 revolutions per minute. Provisions shall be made within the motor to prevent overloading at any point on the pump performance curve in accordance with NFPA 70. Vibration-damping mountings shall support the motor in the mounting bracket.

3.8 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion products or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.9 Dissimilar metals. Metals dissimilar with respect to the galvanic scale shall not be used unless separated by an insulating material which will avoid electrolytic corrosion.

3.10 Workmanship.

3.10.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.10.2 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.10.3 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

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3.10.4 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the casting ability to perform its intended function.

3.10.5 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).
c. Packaging inspection (see 4.5).

4.2.1 First article inspection. First article inspection shall be performed on one pump when a first article is required (see 3.2 and 6.6). This inspection shall include the examination of 4.3 and the tests of 4.4. Failure of the first article to pass the examination or any of the tests shall be cause for rejection. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

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4.2.2 Quality conformance inspection. The quality conformance inspection shall consist of the examination of 4.3, the tests of 4.4, and the packaging inspection of 4.5. When a first article is tested in accordance with the requirements of 4.4, and has passed, the tests of 4.4 need not be performed for the quality conformance inspection. When the tests of 4.4 are to be performed for the quality conformance inspection, they shall be performed on one production unit.

4.3 Examination. Each pump shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests. The first article, when furnished, shall be tested, and any pump failing to pass the following tests, as applicable, shall be rejected. Tests shall be conducted as outlined in the referenced documents and as herein specified.

4.4.1 Hydrostatic test. The pump shall be hydrostatically tested at a pressure equal to 1.5 times the maximum rated head for a period of one minute to determine compliance with 3.6. Tests may be conducted on the completely assembled pump providing no damage to seals, bearings, or other internal components occurs.

4.4.2 Performance test. The pump shall be tested to determine compliance with the requirements of 3.6. Tests shall be conducted and reported in accordance with the Centrifugal Pump Section of the Standards of Hydraulic Institute. Failure of the pump unit to meet any of the performance requirements of this specification shall be cause for rejection.

4.5 Packaging inspection. The preservation, packing, and marking of the pump shall be inspected to the requirements of section 4 of MIL-P-10603. When specified (see 6.2), a preproduction pack inspection shall be performed in accordance with MIL-P-10603.

5. PACKAGING

5.1 Preservation, packing, and marking. The pump shall be packaged, packed, and marked in accordance with MIL-P-10603. The level of preservation and level of packing shall be as specified (see 6.2).

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Pumps covered by this document are intended for use as circulators in forced hot water heating systems and in service water distribution systems where the water temperature for either application does not exceed 225°F.

6.1.1 Selection factors.

a. If three-phase power is not available, single-phase motors may be specified. However, the motor should be rated for the higher system's phase voltage available (see 3.7.6).
b. Fungus resistant varnish conforming to MIL-V-173 should be used to coat electrical components and circuit elements, including terminal and circuit connections, when the pump is to be installed in humid conditions. Components and elements inherently inert to fungi or in hermetically sealed enclosures or current-carrying contact surfaces should not be coated.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Type, size, and service required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.2 and 6.6).
e. Capacity and total dynamic head required (see 3.6).
f. When pump casing and flange are to be other than as specified (see 3.7.1).
g. When material for the impeller is to be other than as specified (see 3.7.2)
h. When the impeller is to be made of brass or bronze (see 3.7.2).
i. When the pump shaft is to be other than as specified (see 3.7.3).
j. When threaded companion flanges are not required (see 3.7.4).
k. When the motor is to be other than as specified (see 3.7.6 and 6.1.1).
l. When a preproduction pack inspection is to be performed (see 4.5).
m. Level of preservation and level of packing required (see 5.1).

6.3 Part or identifying number (PIN). The PIN to be used for pumps applied to this specification are created as follows:

	M16077	XX	X.XX
Specification part number	↓		
Type and service code		↓	
Size code			↓

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6.3.1 Type and service to code identifier. The type and service of the pump is identified by a two-digit identifier as shown in table I. The first character is a single number identifying the type; the second character is a single letter identifying service.

TABLE I. Type and service to code identifier.

	Service A	Service B
Type I	1A	1B
Type II	2A	2B

6.3.2 Size code identifier. The size code identifier is a three-digit number, with decimal point, which corresponds to the size indicated in 1.2.

6.4 Cross-reference of classification. The classification of pumps in this document differs from the previous revision in the following respect:

MIL-P-16077C(YD)	MIL-P-16077D(YD)
CLASSES	
Class 1 - Standard head-capacity.	Class 1 - Deleted.
Class 2 - High head-capacity.	Class 2 - Deleted.

6.5 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27475.1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data shall be delivered by the contractor in accordance with the contract or purchase order requirements.

6.6 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory (see 6.2) as specified in 4.2.1. The first article should consist of one pump. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.7 Subject term (key word) listing.

Circulator pump
Service water distribution system
Hot water heating system
In-line pump

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6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Navy - YD

Review Activity:
DLA - CS

Preparing Activity:
Navy - YD

(Project 4320-N333)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-16077D(YD)	910621

3. DOCUMENT TITLE

PUMPS, CENTRIFUGAL, WATER CIRCULATING, ELECTRIC-MOTOR-DRIVEN

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
COMMANDING OFFICER	(805) 982-5761	551-5761
c. ADDRESS (Include Zip Code) NAVAL CONSTRUCTION BATTALION CENTER CESO (CODE 1564A) PORT HUENEME, CA 93043-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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